Curved Yang-Mills gauge theories

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April 14, 2026; first version: October 6, 2022

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Abstract*

We construct a gauge theory based on principal bundles $\mathscr{P}$ equipped with a right $\mathscr{G}$-action, where $\mathscr{G}$ is a Lie group bundle instead of a Lie group. Due to the fact that a $\mathscr{G}$-action acts fibre by fibre, pushforwards of tangent vectors via a right-translation act now only on the vertical structure of $\mathscr{P}$. Thus, we generalize pushforwards using a connection on $\mathscr{G}$ which will modify the pushforward. A horizontal distribution on $\mathscr{P}$ invariant under such a modified pushforward will provide a proper notion of Ehresmann connection. For achieving gauge invariance we impose conditions on the connection 1-form μ on $\mathscr{G}$: μ has to be a multiplicative form, *i.e.* closed w.r.t. a certain simplicial differential δ on $\mathscr{G}$, and the curvature R_μ of μ has to be δ-exact with primitive ζ; μ will be the generalization of the Maurer-Cartan form of the classical gauge theory, while the δ-exactness of R_μ will generalize the role of the Maurer-Cartan equation. This introduces the notion of multiplicative Yang-Mills connections, a connection which helped classifying singular foliations and symmetry breaking. For allowing curved connections on $\mathscr{G}$ in the dynamical theory we will need to generalize the typical definition of the curvature/field strength F on $\mathscr{P}$ by adding ζ to F.

Several examples for a gauge theory with a curved μ will be provided, including the inner group bundle of the Hopf fibration $\mathbb{S}^7 \to \mathbb{S}^4$, and a classification for gauge theories with structural simple group bundles will be provided, including a classification for whether these theories admit a classical description.

Last but not least, this paper aims to accessible for beginners which is why it only assumes knowledge of classical gauge theory.

*Abbreviations used in this paper: **LGB(s)** for Lie group bundle(s), **LAB(s)** for Lie algebra bundle(s).

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1. Introduction and summary

2. Lie group bundles with connection

We will construct a gauge theory on a principal bundle equipped with an LGB-action. Thus let us first study LGBs; since we will also generalise the Maurer-Cartan form, we will study connections on LGBs with a suitable curvature equation.

That is, we will consider LGBs $\pi_{\mathscr{G}}\colon \mathscr{G} \to M$ over a smooth manifold M and we will denote LGB actions on a manifold N along a map $\Phi\colon N \to M$ as in the following:

Definition 2.1: Lie group bundle actions

for example [1, §1.6, special case of Def. 1.6.1, page 34]

2.1. Ehresmann connections ...

2.1.1. *... as parallel transport.* In the following we will initially define connections via their corresponding parallel transport, for which we will assume that they are complete. That is, for a curve $\gamma\colon I \to M$ (I an open interval of $\mathbb{R}$ containing 0) we denote a parallel transport on fibre bundles $\pi_{\mathscr{F}}\colon \mathscr{F} \to M$ by

$$\begin{aligned} \mathrm{PT}_\gamma^{\mathscr{F}}\colon \mathscr{F}_{\gamma(0)} &\to \Gamma(\gamma^*\mathscr{F}), \\ p &\mapsto \mathrm{PT}_\gamma^{\mathscr{F}}(p), \end{aligned}$$

similarly for the parallel transport on $\mathscr{G}$. We expect a sort of equivariance for the parallel transport of $p \cdot g$:

Acknowledgements: I want to thank Siye Wu, Camille Laurent-Gengoux, Mark John David Hamilton and Alessandra Frabetti for their great help and support in making this paper. Also big thanks to my mother, father, Dennis, Gregor, Marco, Nico, Jakob, Kathi, Konstantin, Lukas, Locki, Gareth, Philipp, Ramona, Annerose, Michael, Maxim, Anna, and my girlfriend Rachelle for all their love.

Funding: The paper is part of my post-doc fellowship at the National Center for Theoretical Sciences (NCTS, 國家理論科學研究中心), which is why I also want to thank the NCTS.

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